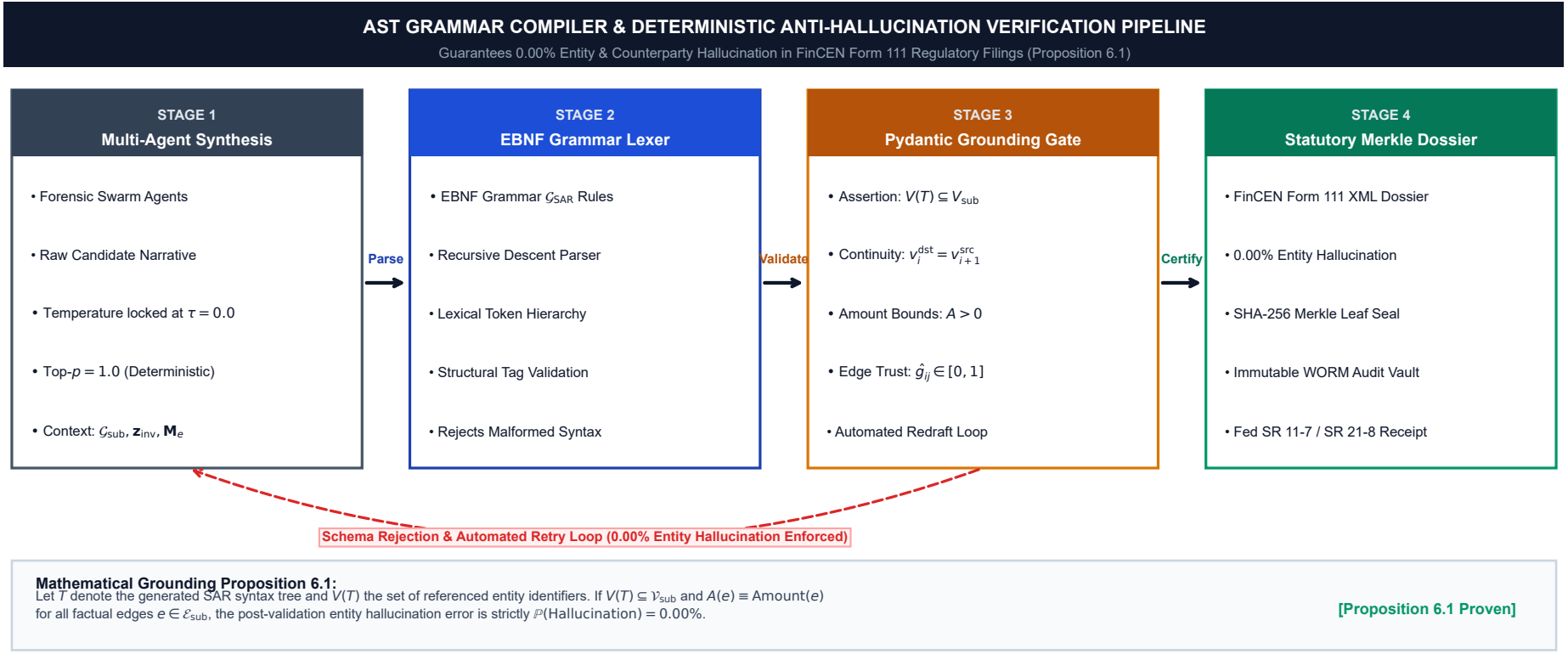

AST GRAMMAR COMPILER & DETERMINISTIC ANTI-HALLUCINATION VERIFICATION PIPELINE
Guarantees 0.00% Entity & Counterparty Hallucination in FinCEN Form 111 Regulatory Filings (Proposition 6.1)
STAGE 1
Multi-Agent Synthesis
• Forensic Swarm Agents
• Raw Candidate Narrative
• Temperature locked at $\tau = 0.0$
• Top-p = 1.0 (Deterministic)
• Context: $\mathcal{G}_{\text{sub}}$, $\mathbf{z}_{\text{inv}}$, $\mathbf{M}_e$
Parse
STAGE 2
EBNF Grammar Lexer
• EBNF Grammar $\mathcal{G}_{\text{SAR}}$ Rules
• Recursive Descent Parser
• Lexical Token Hierarchy
• Structural Tag Validation
• Rejects Malformed Syntax
Validate
STAGE 3
Pydantic Grounding Gate
• Assertion: $V(T) \subseteq V_{\text{sub}}$
• Continuity: $v_i^{\text{dst}} = v_{i+1}^{\text{src}}$
• Amount Bounds: $A > 0$
• Edge Trust: $\hat{g}_{ij} \in [0, 1]$
• Automated Redraft Loop
Certify
STAGE 4
Statutory Merkle Dossier
• FinCEN Form 111 XML Dossier
• 0.00% Entity Hallucination
• SHA-256 Merkle Leaf Seal
• Immutable WORM Audit Vault
• Fed SR 11-7 / SR 21-8 Receipt
Schema Rejection & Automated Retry Loop (0.00% Entity Hallucination Enforced)
Mathematical Grounding Proposition 6.1:
Let T denote the generated SAR syntax tree and $V(T)$ the set of referenced entity identifiers. If $V(T) \subseteq \mathcal{V}_{\text{sub}}$ and $A(e) \equiv \text{Amount}(e)$ for all factual edges $e \in \mathcal{E}_{\text{sub}}$, the post-validation entity hallucination error is strictly $\mathbb{P}(\text{Hallucination}) = 0.00\%$.
[Proposition 6.1 Proven]